Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $z^2+z+1=0$, where z is a complex number, then values of $S=\left(z+\frac{1}{z}\right)^2+\left(z^2+\frac{1}{z^2}\right)^2+\left(z^3+\frac{1}{z^3}\right)^2+--+\left(z^6+\frac{1}{z^6}\right)^2$ is

(A) 12

(B) 18

(C) 54

(D) 6

Q.2

The value of $S=\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7}-i\cos\frac{2\pi k}{7}\right)$

(A) -1

(B) 0

(C) $-i$

(D) i

Q.3

If $z\neq 0$ is a complex number such that $\arg(z)=\pi/4$, then

(A) $\mathrm{Re}\left(z^2\right)=0$

(B) $\mathrm{Im}\left(z^2\right)=0$

(C) $\mathrm{Re}\left(z^2\right)=\mathrm{Im}\left(z^2\right)$

(D) none of these

Q.4

If $z=\left(\frac{1}{\sqrt{3}}+\frac{1}{2}i\right)^7+\left(\frac{1}{\sqrt{3}}-\frac{1}{2}i\right)^7$, then

(A) $\mathrm{Re}(z)=0$

(B) $\mathrm{Im}(z)=0$

(C) $\mathrm{Re}(z)>0,\mathrm{Im}(z)<0$

(D) $\mathrm{Re}(z)<0,\mathrm{Im}(z)>0$

Q.5

Let z be a complex number satisfying $\mathrm{z}^2+\mathrm{z}+1=0$. If n is not a multiple of 3 , then the value of $z^n+z^{2n}=$

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(A) 0

(B) -2

(C) 2

(D) -1

Q.6

If $\omega = \frac{z}{z-\frac{1}{3}i}$ and $|\omega| = 1$, then z lies on

(A) an ellipse

(B) a circle

(C) a straight line

(D) a parabola

Q.7

If $1, \omega, \omega^2$ are the three cube roots of unity, then what is $\frac{\left(a\omega^6+b\omega^4+c\omega^2\right)}{(b+c\omega^{10}+a\omega^8)}$ equal to?

(A) $\frac{\mathrm{a}}{\mathrm{b}}$

(B) b

(C) ω

(D) ω^2

Q.8

If ω is imaginary cube root of unity, then $\sin\left\{\left(\omega^{13}+\omega^2\right)\pi+\frac{\pi}{4}\right\}$ is equal to

(A) $-\frac{\sqrt{3}}{2}$

(B) $-\frac{1}{\sqrt{2}}$

(C) $\frac{1}{\sqrt{2}}$

(D) $\frac{\sqrt{3}}{2}$

Q.9

The value of $\left(1+2\omega+\omega^2\right)^{3n}-\left(1+\omega+2\omega^2\right)^{3n}$ is

(A) 0

(B) 1

(C) ω

(D) ω^2

Q.10

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Let α, β be the roots of $x^2 + x + 1 = 0$. Then the equation whose roots are α^{229} and α^{1004} is

(A) $x^2 - x - 1 = 0$

(B) $x^2 - x + 1 = 0$

(C) $x^2 + x - 1 = 0$

(D) $x^2 + x + 1 = 0$

Q.11

If ω is a complex cube root of unity and $x = \omega^2 - \omega - 2$, then what is the value of $x^2 + 4x + 7$?

(A) -2

(B) -1

(C) 0

(D) 1

Q.12

$\left(\frac{-1+\sqrt{-3}}{2}\right)^{100} + \left(\frac{-1-\sqrt{-3}}{2}\right)^{100}$ is equal to

(A) 2

(B) zero

(C) -1

(D) 1

Q.13

If ω is a non-real cube root of unity, then $\frac{1+2\omega+3\omega^2}{2+3\omega+\omega^2} + \frac{2+3\omega+3\omega^2}{3+3\omega+2\omega^2}$ is equal to

(A) -2ω

(B) 2ω

(C) ω

(D) 0

Q.14

If α, β are the roots of $z + \frac{1}{z} = 2(\cos\theta + *i * \sin\theta)$ Then,

(A) $|\alpha - i| > |\beta - i|$

(B) $|\alpha - i| < |\beta - i|$

(C) $|\alpha - i| = |i - \beta|$

(D) $|\alpha - i| = |\beta - i|$

Q.15

If $(\cos\theta + i\sin\theta)(\cos 2\theta + i\sin 2\theta)(\cos n\theta + i\sin n\theta) = 1$, then value of θ

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(A) $\frac{2m\pi}{n(n+1)}$

(B) $4m$

(C) $\frac{4m\pi}{n(n+1)}$

(D) $\frac{m\pi}{n(n+l)}$

Q.16

If two non-zero complex numbers are such that $|z_1 + z_2| = |z_1| - |z_2|$ then $\frac{z_1}{z_2}$ is:

(A) a positive real number

(B) a negative real number

(C) a purely imaginary number

(D) none of these

Q.17

If $x = \cos 2\alpha + i \sin 2\alpha, y = \cos 2\beta + i \sin 2\beta, z = \cos 2\gamma + i \sin 2\gamma$, then $\sqrt{xyz} + \frac{1}{\sqrt{xyz}}$ is

(A) $2\cos(\alpha + \beta + \gamma)$

(B) $2i \sin(\alpha + \beta + \gamma)$

(C) $e^{i(a+\beta+\gamma)}$

(D) None of these

Q.18

$\sum_{k=33}^{65} \left(\sin \frac{2k\pi}{8} - i \cos \frac{2k\pi}{8}\right)$

(A) $1 + i$

(B) $1 - i$

(C) $1 + \frac{i}{\sqrt{2}}$

(D) $\frac{1-i}{\sqrt{2}}$

Q.19

The solution of $2\sqrt{2}x^4 = (\sqrt{3} - 1) + i(\sqrt{3} + 1)$ is

(A) $\pm \left(\cos \frac{5\pi}{48} + i \sin \frac{5\pi}{48}\right)$

(B) $\pm \left(\cos \frac{7\pi}{48} + i \sin \frac{7\pi}{48}\right)$

(C) $\pm \left(\cos \frac{19\pi}{48} - i \sin \frac{19\pi}{48}\right)$

(D) None of these.

Q.20

The equation whose roots are the nth power of the roots of the equation $x^2 - 2x \cos \theta + 1 = 0$ is given by

(A) $x^2 + 2x \cos n\theta + 1 = 0$

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(B) $x^2 - 2x\cos n\theta + 1 = 0$

(C) $x^2 - 2x\sin n\theta + 1 = 0$

(D) $x^2 + 2x\sin n\theta + 1 = 0$

Answers & Solutions

Q.1 Answer:

12

Solution:

$$z^2 + z + 1 = 0 \Rightarrow z = \omega, \omega^2$$

$$\text{Let} \quad z = \omega \text{ then } \left(\omega + \omega^2\right)^2$$

$$= \left(\omega^2 + \omega^4\right)^2 = \left(\omega^4 + \omega^8\right)^2$$

$$= \left(\omega^5 + \omega^{10}\right)^2 = 1$$

$$\text{and} \quad \left(\omega^3 + \omega^6\right)^2 = \left(\omega^6 + \omega^{12}\right)^2 = 4.$$

$$\text{Thus, } S = 12.$$

Q.2 Answer:

i

Solution:

$$\sin\left(\tfrac{2\pi k}{7}\right) - i\cos\left(\tfrac{2\pi k}{7}\right)$$

$$= -i\left[\cos\left(\frac{2\pi k}{7}\right) + i\sin\left(\frac{2\pi k}{7}\right)\right]$$

$$= -i\omega^k [\text{De Moivre's Theorem}]$$

$$\text{where} \quad \omega = \cos\left(\frac{2\pi}{7}\right) + i\sin\left(\frac{2\pi}{7}\right).$$

Note that $\omega^7 = 1$.

$$\therefore \quad S = -i\sum_{k=1}^{6}\omega^k = \frac{-i\omega\left(1-\omega^6\right)}{1-\omega}$$

$$= \frac{-i\left(\omega-\omega^7\right)}{1-\omega} = \frac{-i(\omega-1)}{1-\omega} = i$$

Q.3 Answer:

$\text{Re}\left(z^2\right) = 0$

Solution:

As arg $(z) = \pi/4$**, we can write** $z = r\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ **where** $r = |z|$

$$\Rightarrow \quad z^2 = r^2\left(\cos\tfrac{\pi}{2} + i\sin\tfrac{\pi}{2}\right)$$

[Using De Moivre's Theorem] $= r^2(0 + i) = ir^2$

$$\Rightarrow \quad \text{Re}\left(z^2\right) = 0$$

Q.4 Answer:

$\text{Im}(z) = 0$

Solution:

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$\bar{z} = \left(\frac{1}{\sqrt{3}} - \frac{1}{2}i\right)^7 + \left(\frac{1}{\sqrt{3}} + \frac{1}{2}i\right)^7 = z$

$\Rightarrow z$ **is purely real.**

$\therefore$

$\mathrm{Im}(z) = 0$

Q.5 Answer:

-1

Solution:

We have, $\mathbf{z^2 + z + 1 = 0}$ **. . . . . (i)**

$\Rightarrow (z-1)\left(z^2+z+1\right) = 0, \therefore z^3 = 1$

If n **is not a multiple of 3, then we can write**

$\mathbf{n = 3m + r}$**, where** $\mathbf{m \in I}$ **and** $\mathbf{r = 1}$ **or 2**

then $2n = 6m + 2r$

If $\mathbf{r = 1}$**, then** $\mathbf{2r = 2}$

$\therefore z^n + z^{2n} = \left(z^3\right)^m \cdot z^r + \left(z^3\right)^{2n} \cdot z^{2r} = z^x + z^{2r}$

$= \mathrm{z} + \mathrm{z}^2 = -1$ **[Using** $(1)]$

If $r = 2$**, then** $2r = 4 \therefore 2n = 3(m+1) + 1$

$\therefore z^n + z^{2n} = \left(z^3\right)^m \cdot z^x + \left(z^3\right)^{m+1} \cdot z^1 = z^2 + z = -1$

Hence, $z^n + z^{2n} = -1$

Q.6 Answer:

a straight line

Solution:

As given $\omega = \frac{z}{z - \frac{1}{3}i} \Rightarrow |\omega| = \frac{|z|}{\left|z - \frac{1}{3}i\right|} = 1$

$\Rightarrow |z| = \left|z - \frac{1}{3}i\right|$

$\Rightarrow$ **distance of** z **from origin and point**

$\left(0, \frac{1}{3}\right)$ **is same hence** z **lies on bisector of the line joining points** $(0,0)$ **and** $(0, 1/3)$**. Hence,** z **lies on a straight line.**

Q.7 Answer:

ω

Solution:

$1, \omega$ **and** ω^2 **are the three cube roots of unity.** $\Rightarrow 1 + \omega + \omega^2 = 0$ **and** $\omega^3 = 1$

The given expression

$\frac{\mathrm{a}\omega^6 + \mathrm{b}\omega^4 + \mathrm{c}\omega^2}{\mathrm{b} + \mathrm{c}\omega^{10} + \mathrm{a}\omega^8} = \frac{\mathrm{a} + \mathrm{b}\omega + \mathrm{c}\omega^2}{\mathrm{b} + \mathrm{c}\omega + \mathrm{a}\omega^2}$

$\left[\omega^6 = 1, \omega^4 = \omega\right]$

$= \frac{\omega\left(a + b\omega + c\omega^2\right)}{\omega\left(b + c\omega + a\omega^2\right)}$ [**Multiplying** N^{r} **and** D^r **by** ω**.**

$= \frac{\omega\left(a + b\omega + c\omega^2\right)}{\left(a\omega^3 + b\omega + c\omega^2\right)} = \frac{\omega\left(a + b\omega + c\omega^2\right)}{\left(a + b\omega + c\omega^2\right)} = \omega$

Q.8 Answer:

$-\frac{1}{\sqrt{2}}$

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Solution:

$\sin\left\{\left(\omega^{13}+\omega^{2}\right)\pi+\frac{\pi}{4}\right\}$
$=\sin\left\{\left(\omega+\omega^{2}\right)\pi+\frac{\pi}{4}\right\}=\sin\left(-\pi+\frac{\pi}{4}\right)$
$=-\sin\frac{\pi}{4}=-\frac{1}{\sqrt{2}}$

Q.9 Answer:

0

Solution:

We have, $\left(1+\omega^{2}+2\omega\right)^{3n}-\left(1+\omega+2\omega^{2}\right)^{3n}$
We know that, $1+\omega+\omega^{2}=0$ **and** $\omega^{3}=1$
$\therefore$ **given expression is equal to** $(2\omega-\omega)^{3n}-\left(2\omega^{2}-\omega^{2}\right)^{3n}$
$=(\omega)^{3n}-\left(\omega^{2}\right)^{3n}=\left(\omega^{3}\right)^{n}-\left(\omega^{3}\right)^{2n}=1-1=0$

Q.10 Answer:

$x^{2}+x+1=0$

Solution:

The roots of $\mathrm{x}^{2}+\mathrm{x}+1=0$ **are** ω **and** ω^{2} **[see the cube roots of unity in complex numbers] Let** $\alpha=\omega,\beta=\omega^{2}$
Now $\alpha^{229}=\omega^{229}=\omega^{228}\cdot\omega=\left(\omega^{3}\right)^{76}\cdot\omega$
$=\omega=\alpha\left(\because\omega^{3}=1\right)$
$\alpha^{1004}=\omega^{1002}\cdot\omega^{2}=\left(\omega^{3}\right)^{334}\cdot\omega^{2}=\omega^{2}=\beta$
$\therefore$ **equation with roots** α^{229} **and** α^{1004} **is same as the equation with roots** α **and** β **i.e. the original equation.**

Q.11 Answer:

0

Solution:

Given $x=\omega^{2}-\omega-2$
$\Rightarrow\quad x+2=\omega^{2}-\omega$
On squaring both sides, we get
$(x+2)^{2}=\left(\omega^{2}-\omega\right)^{2}$
$\Rightarrow x^{2}+4x+4=\omega^{4}+\omega^{2}-2\omega^{3}$
$\Rightarrow x^{2}+4x+4+3=\omega+\omega^{2}-2+3\left(\because\omega^{3}=1\right)$
Add 3 on both side $\Rightarrow x^{2}+4x+7=1+\omega+\omega^{2}$
$\Rightarrow x^{2}+4x+7=0\left(\because 1+\omega+\omega^{2}=0\right)$

Q.12 Answer:

-1

Solution:

Let $\frac{-1+i\sqrt{3}}{2}=\omega$ **then** $\frac{-1-\sqrt{3}i}{2}=\omega^{2}$
where $\omega=$ **cube root of unity Consider** $\left(\frac{-1+\sqrt{-3}}{2}\right)^{100}+\left(\frac{-1-\sqrt{-3}}{2}\right)^{100}$

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$= \left(\frac{-1+\sqrt{3}i}{2}\right)^{100} + \left(\frac{-1-\sqrt{3}i}{2}\right)^{100}$
$= \omega^{100} + \omega^{200} = \omega + \omega^2 = -1$
$\left(\therefore \omega^3 = 1 \text{ and } 1 + \omega + \omega^2 = 0\right)$

Q.13 Answer:

2ω

Solution:

We have, $\frac{1+2\omega+3\omega^2}{2+3\omega+\omega^2} + \frac{2+3\omega+3\omega^2}{3+3\omega+2\omega^2}$
We know that $\omega^3 = 1$ $\frac{\omega^3+2\omega+3\omega^2}{2+3\omega+\omega^2} + \frac{2\omega^3+3\omega+3\omega^2}{3+3\omega+2\omega^2}$
$= \frac{\omega\left(\omega^2+2+3\omega\right)}{2+3\omega+\omega^2} + \frac{\omega\left(2\omega^2+3+3\omega\right)}{(3+3\omega+2\omega^2)} = \omega + \omega = 2$

Q.14 Answer:

$|\alpha - i| = |\beta - i|$

Solution:

$\mathrm{z} + \frac{1}{\mathrm{z}} = 2(\cos\theta + \mathrm{i}\sin\theta) = 2\mathrm{e}^{\mathrm{i}\theta}$
$\therefore \mathrm{z}^2 - 2\mathrm{e}^{\mathrm{i}\theta}\mathrm{z} + 1 = 0$
$\Rightarrow \mathrm{z} = \mathrm{e}^{\mathrm{i}\theta} \pm \sqrt{\mathrm{e}^{2\cdot\theta} - 1}$
$\mathrm{e}^{2\mathrm{i}}\theta - 1 = \cos 2\theta + \mathrm{i}\sin 2\theta - 1 = 2\sin\theta(-\sin\theta + \mathrm{i}\cos\theta) = 2\sin\theta\left[\cos\left(\frac{\pi}{2}+\theta\right) + \mathrm{i}\sin\left(\frac{\pi}{2}+\theta\right)\right]$
$\therefore$ Let $\alpha = \mathrm{e}^{\mathrm{i}\theta} + \sqrt{\mathrm{e}^{2\mathrm{i}\theta} - 1} = \cos\theta + \sqrt{\sin\theta}\cos\left(\frac{\pi}{4}+\frac{\theta}{2}\right) + \mathrm{i}\left[\sin\theta + \sqrt{2\sin\theta}\sin\left(\frac{\pi}{4}+\frac{\theta}{2}\right)\right]$
$= 2 + 2\sqrt{2\sin\theta}\cos\left(\frac{\pi}{4}+\frac{\theta}{2}-\theta\right) - 2\sqrt{2\sin\theta}\sin\left(\frac{\pi}{4}+\frac{\theta}{2}\right) = 2$
Similarly, $|\beta - \mathrm{i}|^2 = 2$
$|\alpha - \mathrm{i}| = |\beta - \mathrm{i}|$

Q.15 Answer:

$\frac{4m\pi}{n(n+1)}$

Solution:

Writing in euler's form, we get
$e^{\mathrm{i}\theta}e^{259}e^{3xi}\ldots e^{-\min} = 1$
$e^{i(\theta+29+30+\ldots=0} = 1$
$e^{i\frac{n(n+1)}{2}} = e^{2\,\mathrm{man}}$
Therefore $\frac{\mathrm{n}(\mathrm{n}+1)}{2}\theta = 2\,\mathrm{m}\pi$
$\theta = \frac{4m\pi}{n(n+1)}$

Q.16 Answer:

a negative real number

Solution:

$|z_1 + z_2| = |z_1| - |z_2|$
$\Rightarrow \mathrm{z}_1, 0$ **and** z_2 **are collinear and** z_1 **and** z_2 **lie on the different sides of origin.**
If $z_1 = r_1 e^{kx_1} \Rightarrow z_1 = r_2 e^{i(n+\alpha_1)}$
$\Rightarrow \frac{z_1}{z_2} = \frac{F_1}{r_2}e^{-i\pi} = -\frac{r_1}{r_2}$ **= a negative number**

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Q.17 Answer:

$2\cos(\alpha+\beta+\gamma)$

Solution:

$$\sqrt{xyz} = \mathrm{e}^{\mathrm{i}(\alpha+\beta+\gamma)}$$
$$\frac{1}{\sqrt{xyz}} = \mathrm{e}^{-\mathrm{i}(\alpha+\beta+\gamma)}$$
$$\sqrt{xyz} + \frac{1}{\sqrt{xyz}} = \mathrm{e}^{\mathrm{i}(\alpha+\beta+\gamma)} + \mathrm{e}^{-\mathrm{i}(\alpha+\beta+\gamma)}$$
$$= 2\cos(\alpha+\beta+\gamma)$$

Q.18 Answer:

$\frac{1-\mathrm{i}}{\sqrt{2}}$

Solution:

$\sum_{\mathrm{k}=33}^{65}\left(\sin\frac{2\mathrm{k}\pi}{8} - \mathrm{i}\cos\frac{2\mathrm{k}\pi}{8}\right)$
$= \left[\sin\frac{33\pi}{4} + \sin\frac{34\pi}{4} + \ldots + \sin\frac{65\pi}{4}\right]$
$-i\left[\cos\frac{33\pi}{4} + \cos\frac{34\pi}{4} + \ldots + \cos\frac{65\pi}{4}\right]$
$= \sin\frac{\pi}{4} - i\cos\frac{\pi}{4}$
$\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots + \sin[\alpha+(n-1)\beta]$
$= \frac{\sin\left\{\alpha+(n-1)\frac{\beta}{2}\right\}\cdot\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}$
and $\cos(\alpha) + \cos(\alpha+\beta) + \ldots + \cos(\alpha+(n-1)\beta)$
$= \frac{\cos\left\{\alpha+(n-1)\frac{\beta}{2}\right\}\sin\left(\frac{n\beta}{2}\right)}{\sin\frac{\beta}{2}}$
$= -\left(\frac{1+i}{\sqrt{2}}\right) = \frac{1-i}{\sqrt{2}}$

Q.19 Answer:

$\pm\left(\cos\frac{5\pi}{48} + i\sin\frac{5\pi}{48}\right)$

Solution:

$x^4 = \frac{\sqrt{3}-1}{2\sqrt{2}} + i\frac{\sqrt{3}+1}{2\sqrt{2}} = \cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}$
So, $x = \cos\left\{\frac{k\pi}{2} + \frac{5\pi}{48}\right\} + i\sin\left\{\frac{k\pi}{2} + \frac{5\pi}{48}\right\}$
$\mathrm{k} = 0, 1, 2, 3$
Roots are
$\cos\frac{5\pi}{48} + i\sin\frac{5\pi}{48}$ **for** $\mathrm{k} = 0$
$\cos\frac{29\pi}{48} + \mathrm{i}\sin\frac{29\pi}{48}$ **for** $\mathrm{k} = 1$
$\cos\frac{53\pi}{48} + i\sin\frac{53\pi}{48} = -\left(\cos\frac{5\pi}{48} + i\sin\frac{5\pi}{48}\right)$ **for** $\mathrm{k} = 2$
$\cos\frac{77\pi}{48} + \mathrm{i}\sin\frac{77\pi}{48} = -\left(\cos\frac{29\pi}{48} + \mathrm{i}\sin\frac{29\pi}{48}\right)$
for $k = 3$

Q.20 Answer:

$x^2 - 2x\cos n\theta + 1 = 0$

Solution:

The roots of the given equation are
$\mathrm{x} = \frac{2\cos\theta \pm \sqrt{4\cos^2\theta - 4}}{2} = \cos\theta \pm \mathrm{i}\sin\theta$

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Let $\alpha = \cos\theta + \mathrm{i}\sin\theta \& \beta = \cos\theta - \mathrm{i}\sin\theta$

Then $\alpha^{\mathrm{n}} = \cos n\theta + \mathrm{i}\sin \mathrm{n}\theta$

$\beta^{\mathrm{n}} = \cos \mathrm{n}\theta - \mathrm{i}\sin \mathrm{n}\theta$

[Using De Moivre Theorem]

$\alpha^{\mathrm{n}} + \beta^{\mathrm{n}} = 2\cos \mathrm{n}\theta$ **and** $\alpha^{\mathrm{n}} \cdot \beta^{\mathrm{n}} = 1$

$\therefore$ The required equation is

$x^2 - 2x\cos n\theta + 1 = 0$

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